Ricardo (Zhicong) Xie

Systems / Project Engineering Intern | Requirements, Risk and Technical Coordination

Auckland, New Zealand | 02902729390 | zxie588@aucklanduni.ac.nz | Portfolio: https://zxie588-alt.github.io | GitHub: https://github.com/zxie588-alt

Valid NZ student visa | No sponsorship required | Full-time 17 Nov 2026 - late Feb 2027

PROFILE

- Mechanical engineering master's student with team-lead project experience in requirements, stakeholder analysis, trade studies, risk logic and verification planning.
- Best suited to project engineering, systems engineering, infrastructure, smart manufacturing and technical coordination internships where technical decisions need to be organised and explained.
- Project screenshots and source links: https://zxie588-alt.github.io

WORK RIGHTS AND AVAILABILITY

- Valid NZ student visa. No sponsorship required.
- Full-time available: 17 Nov 2026 - late Feb 2027 (university break).
- Part-time eligible: up to 25 hours/week during semester. Auckland based; open to relocation for suitable roles.

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study and project work in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods.

Relevant coursework: Engineering Methods, Advanced Industry 4.0 Smart Manufacturing, Manufacturing and Industrial Systems, Research Project

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PROJECT WORK

Orbital AI Data Centre Feasibility Study - Systems Lead

Stakeholders, requirements, ConOps, trade studies, risk, verification

- Led the systems-engineering workstream for a 5-person team concept covering a 24-satellite LEO AI data-centre architecture.
- Set up the stakeholder map, requirements hierarchy, ConOps, architecture options, trade-study criteria, risk register and verification gates.
- Delivered a feasibility-report structure with trade-off matrices, risk logic and verification-gate framework across orbit/coverage, power, thermal rejection, communications, launch mass, cost and sustainability.

NZ Residential MVHR / ERV Ventilation Concept

SolidWorks, AutoCAD/DXF, HVAC sizing, NZBC G4/H1 context, commissioning plan

- Developed a New Zealand residential MVHR/ERV ventilation concept aligned with local building-code requirements.
- Modelled the MVHR unit, supply/extract/outdoor/exhaust ducts, diffusers, grilles and hangers in SolidWorks; exported DXF, STEP, STL, PNG and a component list for review.
- Sized the system for 60 L/s balanced supply/extract in normal operation and 85 L/s boost extract mode, calculating external static pressure targets of about 80 Pa and 120 Pa.
- Checked the design against NZBC G4/H1 and Healthy Homes ventilation context; developed a commissioning test plan covering CO2, humidity, airflow and fan-power measurements.

Ruggedized Outdoor IoT Sensor Module

SolidWorks, enclosure design, DFM notes, Keil C, concept validation

- Designed a SolidWorks enclosure concept with screw bosses, gasket stack-up, PCB and battery placeholders, cable/strap mounts and service access.
- Applied DFM principles to improve enclosure features for CNC machining discussion and 3D-print prototype planning.
- Added FEA-style screening for corner load, strap lug and screw boss regions; kept an open test list for sealing, impact, thermal exposure and firmware validation.
- Built concept firmware in Keil C with mock sensor readout, battery monitoring and a simple module state machine; documented the electronics and firmware architecture in the portfolio.

Manufacturing Workshop Training

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared 2D AutoCAD/DXF geometry for laser cutting and engraving, then compared drawing intent with the physical cut plate.
- Observed and practised CNC machining workflow including workholding, spindle/tool motion, cutting fluid, chip formation and basic workshop safety protocols aligned with general WorkSafe NZ expectations.
- Reviewed machined gear and shaft components to connect CAD features with real manufacturing constraints such as tolerances, edge quality and tool access.

TECHNICAL SKILLS

- Proficient in: stakeholder mapping, requirements hierarchy, ConOps, requirements traceability, trade studies, risk register and verification gates
- Proficient in: team alignment, assumption tracking, decision logs, technical integration, review preparation and evidence-based documentation
- Familiar with: HVAC layout, manufacturing constraints, simulation checks, mechatronic sensor systems, SolidWorks, AutoCAD/DXF, Abaqus and ANSYS Fluent/CFX

LEADERSHIP AND COMMUNICATION

- Led team-level integration in the orbital data-centre project and commonly take the organiser role in group work.
- Skilled in technical visual communication and structured documentation; experienced in presenting engineering analysis clearly and producing evidence-based project deliverables.
- Professional working proficiency in English; native Mandarin Chinese.